

Ranim BEJI

✉ ranymbeji8@gmail.com

in <https://www.linkedin.com/in/ranim-beji/>

🌐 <http://ranymbeji8.wixsite.com/2001>



Education

- 2022 – 2024 **Master of Research (MRes), Higher Institute of Environmental Sciences and Technology, Borj Cedria, Tunisia** in Energy and Processes.
Thesis: *Electrical and Thermal Performance Analysis of a Sensor Solar Hybrid PV/T Air*
Grade: Graduated with Honors.
<http://ranymbeji8.wixsite.com/2001/projets>
- 2019 – 2022 **License in Maintenance of Energy and Fluid Systems, Higher Institute of Environmental Sciences and Technology, Borj Cedria, Tunisia.**
Thesis: *Study of a 200 kWc Floating Photovoltaic Plant.*
Grade: Graduated with High Honors.
- 2018 – 2019 **Bachelor, Mohamed Brahmi High School, Tunisia.**
Specialty: *Technical Sciences.*

Research and Professional Experience

- 8 Months **Thermal Processes Laboratory (TPL), Energy Research and Technology Center (CRTEn), Borj Cedria Technopole,**
- Develop a comprehensive physical model of the solar cogeneration process using a hybrid PV/T air collector
 - perform numerical simulations to analyze the operational behavior of the studied system
 - Design and implement a data acquisition system for computer-based management of the process, including experimental measurements
 - Evaluate and interpret results across various operating modes of the PV/T air system
- 6 months **Tunisian Company of Electricity and Gas (STEG),**
- End of study project internship: study of a 200 KWc floating photovoltaic power plant through ground PV feasibility study
 - Design of the floating structure to support the solar panels
 - evolution of expected energy production
- <http://ranymbeji8.wixsite.com/2001/projets>

Research and Professional Experience (continued)

- 2 Months

The National Company of Oil Distribution (AGIL),

- Supported photovoltaic installations at two service stations
 - Assisted in oil supply management and contract negotiations
 - Optimization of operations and improvement of energy efficiency
 - Analyzed energy data and proposed operational improvements
- 3 months

Tunisia's food industries (IAT),

- Researched optimization of industrial energy processes
 - Participate in the implementation of energy efficiency and cost reduction solutions
 - Conducted data analysis to enhance system performance

Skills

- Languages:

French (C1: fluent in reading, writing, and speaking), English (B2: upper-intermediate, actively improving through advanced coursework), Arabic (native).
- Programming and Simulation:

Matlab, Python, FORTRAN, COMSOL Multiphysics, PVsyst, TRNSYS, EnergyPlus.
- Engineering Design and Modeling:

AutoCAD, SolidWorks, SketchUp, Arduino.
- Data Acquisition and Analysis:

LABVIEW, experimental data processing, numerical simulation.
- Scientific Communication:

L^AT_EX typesetting, academic research, technical writing, consultation.

Miscellaneous Experience

Certification

- 2021

Certified ISO 14001.

Certified ISO 50001.
- 2024

Certified Python.
- 2025 (Expected)

English Language Proficiency (C1-Level Coursework in Progress)